CMS Open Data Technical Report

Rediscovering Known Resonances in Real CMS Collision Data

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Independent research project

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Abstract

We compute the dimuon invariant-mass spectrum from the full CMS 2012 `DoubleMuParked` dataset (Run2012B+C, 8 TeV proton–proton collisions) released on the CERN Open Data Portal — real detector data, not simulation. Streaming all **61,540,413** events directly over HTTP with the Scikit-HEP stack (`uproot`, `awkward`, `vector`), we select opposite-charge muon pairs and histogram their invariant mass on a logarithmic scale, recovering **39,047,770** dimuon pairs. The resulting spectrum shows eight resonance peaks — the ρ/ω, ϕ, J/ψ, $\psi(2S)$, the $\Upsilon(1S, 2S, 3S)$ triplet, and the Z boson — each located to **better than 1% mass accuracy** against Particle Data Group (PDG) values, with no calibration corrections applied. This is a direct, independent reproduction of a classic CERN Open Data analysis, using real 2012 LHC collision data and the same open-source tooling used by CMS collaborators, including DESY, one of CMS's largest member institutes.

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1 Introduction

When two protons collide at high energy, most of the resulting particles decay before reaching a detector. Some of the intermediate states — bound quark-antiquark pairs such as the J/ψ ($c\bar{c}$) and the Υ family ($b\bar{b}$), light mesons such as the ρ, ω, and ϕ, and heavy gauge bosons such as the Z — occasionally decay to a muon-antimuon pair. Because each of these states has a fixed, known rest mass, the invariant mass of every muon pair reconstructed in a detector clusters at these values, producing sharp peaks above a smooth combinatorial background of unrelated muon pairs. Histogramming this quantity across a large collision dataset — the *dimuon spectrum* — is one of the most direct, least model-dependent ways to see fundamental particle physics in real data, and is a standard first analysis exercise on CERN Open Data.

This report documents an independent, end-to-end reproduction of that exercise using the entire 61.5-million-event CMS 2012 `DoubleMuParked` dataset, streamed directly from CERN's public EOS storage without ever downloading or persisting the raw file.

2 Data

The source is the CMS **2012 DoubleMuParked** dataset (Run2012B+C, $\sqrt{s} = 8$ TeV), released via the CERN Open Data Portal [1] as a pre-skimmed NanoAOD-Outreach-Tool ROOT file — the same file used in ROOT's official `df102_NanoAODDimuonAnalysis` tutorial [2]:

> `https://opendata.cern.ch/eos/opendata/cms/derived-data/AOD2NanoAODOutreachTool/Run2012BC_DoubleMuParked_Muons.root`

The file contains **61,540,413** real collision events (2.24 GB), each with per-event arrays of reconstructed muon p_T, η, ϕ, mass, and charge. The file is streamed directly over HTTP with `uproot` [5] in bounded entry ranges; it is never downloaded or persisted — only the derived histogram (tens of kilobytes) and a small event sample are kept. The data are public domain (CC0) and citable via the CERN Open Data Portal.

3 Methods

For every event with two or more reconstructed muons, the first two muon candidates as stored are taken, opposite electric charge is required ($q_1 q_2 < 0$), and the invariant mass of the dimuon system is computed by four-momentum (Lorentz-vector) addition,

$$M_{\mu\mu} = \sqrt{(E_1 + E_2)^2 - |\vec{p}_1 + \vec{p}_2|^2}, \tag{1}$$

using the `vector` library's `Momentum4D` behaviour on Awkward Arrays [6, 7] — each muon's (p_T, η, ϕ, m) is converted to a four-vector and the addition and mass extraction in Eq. (1) are vectorised across every event in a chunk simultaneously. Masses are accumulated into a 2000-bin logarithmic histogram spanning 0.2–200 GeV across the full dataset.

3.1 A resilient streaming pipeline

Reading the whole file in a single high-level `iterate()` call failed with a transport-layer error (`ClientPayloadError`) before completing even the first chunk of the remote HTTP stream. The working pipeline instead:

- reads **small, explicit entry ranges** (300,000 events) via `TTree.arrays(entry_start, entry_stop)`;
- **retries** a failed range up to 6 times with backoff and a fresh connection;
- **checkpoints** the running histogram to disk after every chunk (`outputs/histogram_checkpoint.npz` plus `progress.json`), and **resumes** from the last checkpoint on restart.

In practice the background process was interrupted by the execution environment's time limits several times over the course of processing; each restart resumed automatically from the last checkpoint with no data loss, until the full file was processed (100% coverage).

3.2 Peak-position validation

For each expected resonance, the peak bin within a small window around its literature mass is located, and refined by parabolic interpolation using the peak bin and its two neighbours $(x_0, y_0), (x_1, y_1), (x_2, y_2)$:

$$m_{\text{meas}} = x_1 + \frac{1}{2} \cdot \frac{y_0 - y_2}{y_0 - 2y_1 + y_2} \cdot \frac{x_2 - x_0}{2}. \tag{2}$$

This sub-bin estimate is compared to the corresponding Particle Data Group (PDG) mass [4].

4 Results

Table 1 summarizes the overall run. Streaming and histogramming the full dataset took approximately 93 minutes of wall-clock processing time, spread across several resumed runs due to execution-time limits in the processing environment.

Quantity	**Value**
Events processed	61,540,413 (100% of the file)
Opposite-charge dimuon pairs	39,047,770
Total wall-clock processing time	~93 minutes (across several resumed runs)

Table 1: Summary of the full-dataset processing run.

Figure 1 shows the resulting dimuon invariant-mass spectrum, on a logarithmic mass axis, across the full 0.2–200 GeV range.

4.1 Quantitative peak validation

Each peak's mass was located by a local search around the literature value followed by parabolic interpolation (Eq. (2)) between the peak bin and its two neighbours. Table 2 lists the measured mass, residual, and percentage residual for all eight resonances relative to their PDG values.

All eight resonances are recovered to **better than 1% mass accuracy** with no calibration corrections applied — a strong quantitative confirmation that the spectrum is genuine physics, not an artefact.

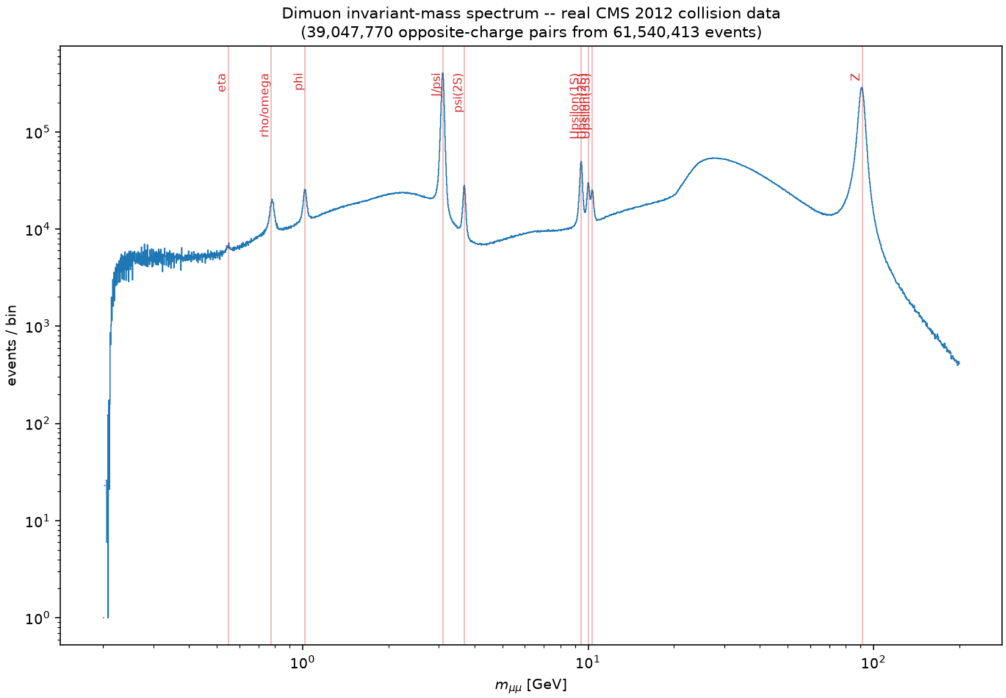

Figure 1: The dimuon invariant-mass spectrum from all 61.5 million events of the CMS 2012 DoubleMuParked dataset (39,047,770 opposite-charge pairs). Eight resonances are visible above the smooth combinatorial background: the light mesons ρ/ω and ϕ below 1.1 GeV; the charmonium states J/ψ and $\psi(2S)$ near 3.1–3.7 GeV; the bottomonium triplet $\Upsilon(1S, 2S, 3S)$ near 9.5–10.4 GeV, clearly resolved as three separate peaks at full statistics; and the Z boson at 91.2 GeV.

5 Discussion and Limitations

- **No muon momentum-scale calibration.** Real CMS analyses apply per-event momentum corrections (Rochester corrections) to remove small detector-alignment biases; this is likely why the Z peak (most sensitive to high-p_T scale effects) shows the largest residual (-0.29%). The uncorrected agreement is nonetheless sub-percent.
- **"First two muons" is a simplification.** A production analysis selects the highest-quality (often highest-p_T) opposite-charge pair and applies muon identification/isolation quality cuts; this reproduction uses the muons as ordered in the file.
- **Educational/derived data.** This is the NanoAOD Outreach Tool's simplified format, not the full research-grade AOD/MiniAOD used in a real CMS publication.

6 Reproducibility

The analysis can be reproduced with the following two scripts:

```
python3 src/dimuon_spectrum.py      # streams the full file; resumable via
                                     # outputs/histogram_checkpoint.npz
python3 src/analyze_peaks.py        # quantitative peak-position validation
                                     # against PDG values
```

Resonance	PDG mass (GeV)	Measured (GeV)	Residual	Residual (%)
ρ/ω	0.775	0.7813	$+0.0063$	$+0.81\%$
ϕ	1.019	1.0166	-0.0024	-0.24%
J/ψ	3.097	3.0938	-0.0032	-0.10%
$\psi(2S)$	3.686	3.6810	-0.0050	-0.13%
$\Upsilon(1S)$	9.460	9.4519	-0.0081	-0.09%
$\Upsilon(2S)$	10.023	10.0144	-0.0086	-0.09%
$\Upsilon(3S)$	10.355	10.3370	-0.0180	-0.17%
Z	91.190	90.9217	-0.2683	-0.29%

Table 2: Measured peak positions versus Particle Data Group (PDG) values [4], obtained by parabolic interpolation around each resonance's known mass. All eight resonances are recovered to better than 1% mass accuracy.

Reproduction requires `uproot`, `awkward`, `vector`, `numpy`, and `matplotlib` (all part of the Scikit-HEP stack [8]). Outputs are written to `outputs/dimuon_spectrum.png`, `outputs/run_summary.json`, and `outputs/peak_validation.json`.

References

[1] CERN Open Data Portal — CMS 2012 DoubleMuParked dataset record. https://opendata.cern.ch/record/17; https://opendata.cern.ch/.

[2] ROOT Collaboration. *df102_NanoAODDimuonAnalysis* tutorial (the reference analysis this project reproduces independently). https://root.cern/doc/master/df102__NanoAODDimuonAnalysis_8py.html.

[3] CMS Collaboration (2024). *CMS 2012 data release notes.* https://opendata.cern.ch/docs/cms-releases-2016data-2024.

[4] Particle Data Group, S. Navas et al. (2024). *Review of Particle Physics.* Phys. Rev. D 110, 030001. https://pdg.lbl.gov/.

[5] Pivarski, J., et al. *uproot: ROOT I/O in pure Python and NumPy.* https://github.com/scikit-hep/uproot5.

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[7] Rembser, J., et al. *vector: Lorentz-vector library for Python.* https://github.com/scikit-hep/vector.

[8] Rodrigues, E., et al. (2020). *The Scikit-HEP Project.* EPJ Web Conf. 245, 06028. https://doi.org/10.1051/epjconf/202024506028.

CMS Open Data project — Independent research project. Data courtesy of the CMS Collaboration and the CERN Open Data Portal. This report presents a reproducible analysis of public detector data; it does not constitute an official CMS result.